MSc in Artificial Intelligence and Data Science Assignment
Understanding AI (772237_A24_T2)

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Exercise 1 — Analysing Second Hand Car Sales Data

1. Introduction

This project examines a dataset of 55,000 second-hand car sales in the UK, documenting vital vehicle details: manufacturer, model, engine size, fuel type, year of manufacture, mileage, and sale price. The key purpose is to design and contrast supervised learning models for precisely estimating car prices, spanning from single-feature linear and polynomial regressions to multi-feature linear and polynomial models, tree-based methods, and artificial neural networks (ANNs). Further, unsupervised learning techniques, including k-Means clustering and other clustering algorithms are implemented to detect trends and groupings within the data frame. By precisely reviewing these models, the purpose of the research is to identify the major elements affecting price variations and to establish strong forecasting techniques. This study presents insights into price drivers and data structure, directing strategic decisions for reliable vehicle assessment and discovering implicit clusters in the second-hand car market.

Table 1. Dataset summary:

Attribute	Type	Unique Values / Categories	Mean / Mode	Std / Range	Min	Max
Manufacturer	object	5	Ford	-	-	-
Model	object	15	Mondeo	-	-	-
Engine size	float64	-	1.77	0.73	1.0	5.0
Fuel type	object	3	Petrol	-	-	-
Year of manufacture	int64	-	2004.21	9.65	1984	2022
Mileage	int64	-	112497.32	71632.52	630	453537
Price	int64	-	13828.9	16416.68	76	168081
Model	object	15	Mondeo	-	-	-

2. Data Preprocessing

The dataset was initially downloaded and analysed, comprising 50,000 records with columns illustrating manufacturer, model, engine size, fuel type, year of manufacture, mileage, and price. Data types were inspected and revised, with discreate variables (Manufacturer, Model, Fuel type) suitably converted to category types, and numerical features (Engine size, Year of manufacture, Mileage, Price) confirmed as numeric, ensuring adaptability to further modelling.

Each attribute was reviewed and found to be having no missing values, making pre-processing process much simple. Repeated records were found (12 rows) and eradicated, converting the dataset to 49,988 unique entries, as a result the training model is prevented from being biased due to redundant information**.**

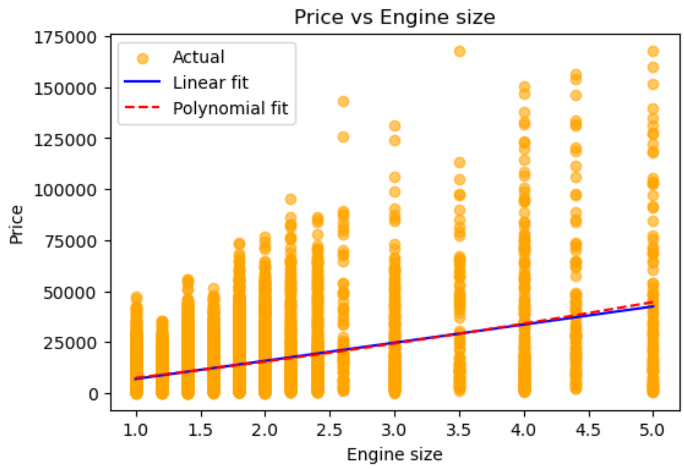
Plausibility tests were performed to verify the cohesion of the data: Year of manufacture was confirmed to fall between 1886 and the current year, Engine size within 0.6–8.0 litres, and Mileage and Price were positive. These tests confirmed that apparent data-entry errors were eradicated before conducting the analysis.

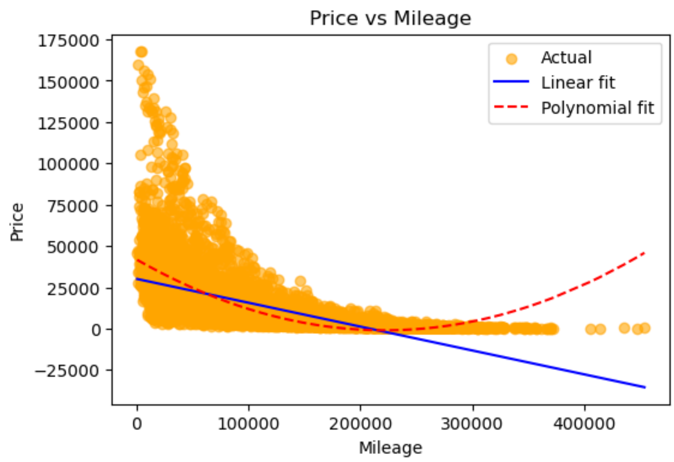
Subsequent to these steps, the dataset was claimed to be organised and ready for modelling. Complementary feature engineering, encoding, and scaling would be conducted in succeeding phases to improve forecasting capability for both supervised and unsupervised learning models.

3. Baseline & Single-feature Regression

To establish reference points for predictive performance, **baseline models** were first implemented: the **mean predictor** (predicting the average price for all cars) and the **median predictor**, providing minimal benchmarks for evaluation.

Single-feature regressions were conducted to evaluate the predictive power of individual numerical variables: **Engine size**, **Year of manufacture**, and **Mileage**. Both **linear regression** and **polynomial regression (degrees 2 and 3)** were fitted.





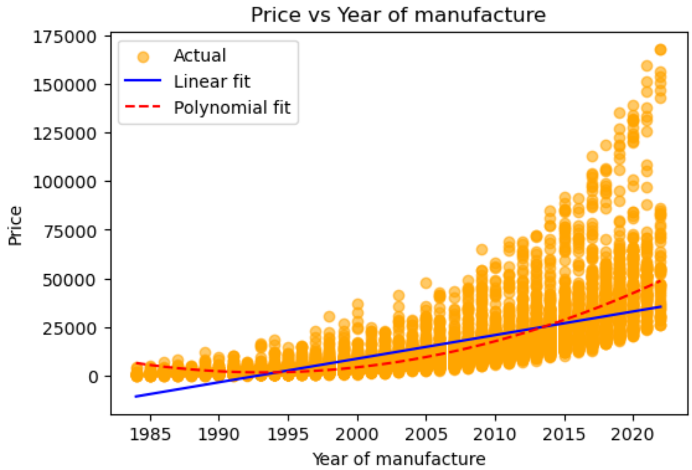


Figure 1. Scatter plots showing linear and polynomial fits for each numerical feature, highlighting differences in predictive performance.

Model performance metrics are summarized in Table 1. **Year of manufacture (Polynomial)** exhibited the **highest R^2 (0.615) and lowest RMSE (10,206.43)**, indicating it is the most informative single predictor. In contrast, **Engine size (Linear/Polynomial)** had the **lowest R^2 (~0.156) and highest RMSE (~15,113)**, making it the least predictive. Polynomial regression slightly improved performance for Year of manufacture and Mileage, but had minimal effect for Engine size.

Residual patterns confirm these observations: errors are smallest for Year of manufacture and largest for Engine size, supporting the quantitative evaluation.

Table 2. Performance metrics comparing linear and polynomial regressions for single-feature models, identifying Year of manufacture as the strongest predictor.

Feature	Linear R2	Linear RMSE	Polynomial R2	Polynomial RMSE
Engine size	0.156	15113.00	0.157	15106.61
Year of manufacture	0.516	11441.83	0.615	10206.43
Mileage	0.404	12698.14	0.523	11364.15

4. Multiple-feature Regression

Building on the single-feature analysis, a **multiple-feature regression model** was implemented using the numerical variables **Engine size, Year of manufacture, and Mileage**. Both **linear** and **polynomial regression** models were fitted to assess the combined predictive power of these features.

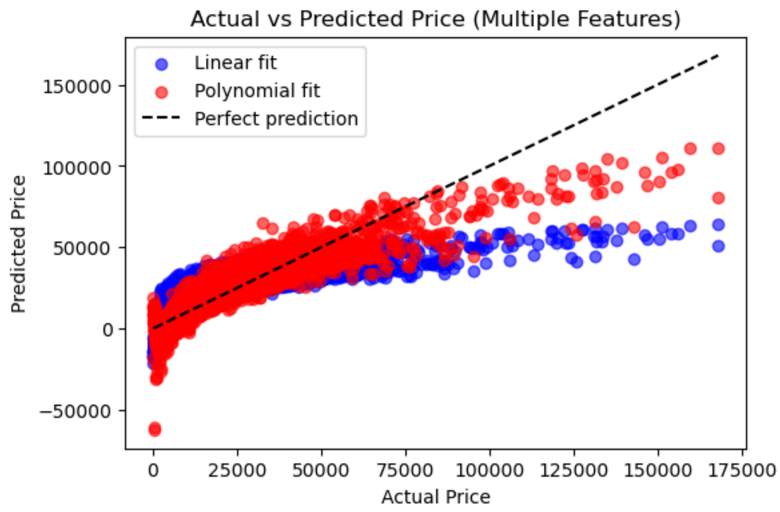


Figure 2. Scatter plot comparing actual and predicted prices for multiple-feature regression, highlighting superior fit of the polynomial model.

Performance metrics are summarized in Table 2. The linear model obtained $R^2 = 0.681$ and RMSE = 9,286.01, exhibiting a significant enhancement over single-feature regression. The polynomial model further improved estimated precision, achieving $R^2 = 0.862$ and RMSE = 6,102.31, indicating substantial reduction in forecasting error. This demonstrates that combining multiple numerical variables captures more variance in car prices than individual features alone.

Residual patterns confirm the improved fit of the polynomial model, with errors substantially reduced across the price range. The results quantitatively highlight the benefit of including multiple features: predictive performance is significantly enhanced, particularly for high-priced vehicles where single-feature models underperformed.

Table 3. Performance metrics for multiple-feature regression, showing polynomial regression substantially outperforms linear regression.

Features	Linear R2	Linear RMSE	Polynomial R2	Polynomial RMSE
Engine size + Year of manufacture + Mileage	0.681	9286.01	0.862	6102.31

5. Regression with Categorical Variables

To integrate both qualitative and quantitative data, a Random Forest Regressor was fitted on the entire dataset. Categorical attributes (Manufacturer, Model, Fuel type) were encoded via one-hot encoding, while quantitative variables were maintained in their raw scales. This integration enabled the model to adaptively represent complex, non-linear interactions between attributes without the rigid linearity assumptions of prior regression models.

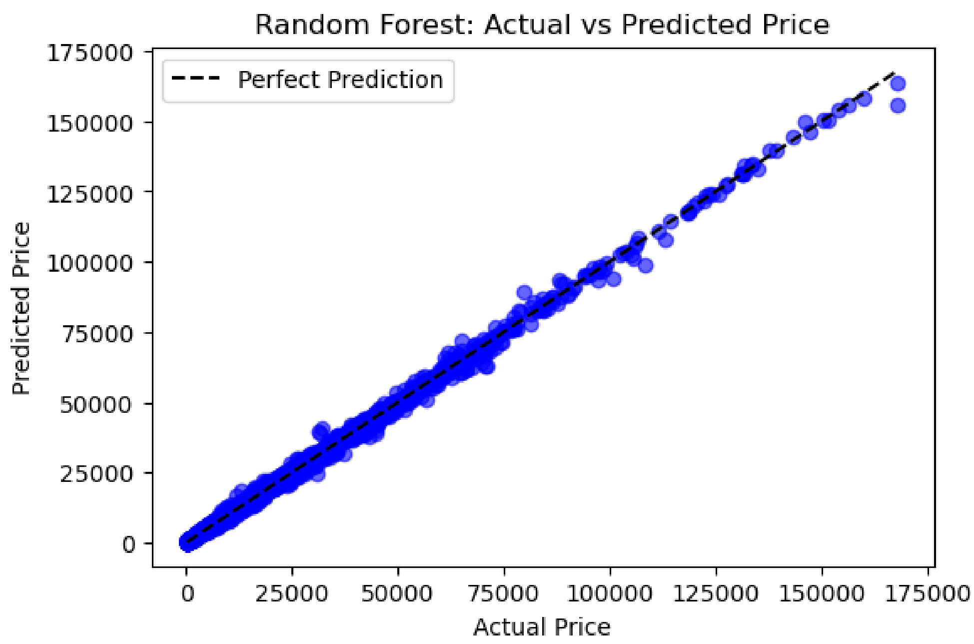


Figure 3. Actual vs predicted prices for the Random Forest model, showing near-perfect alignment with the reference line.

Figure 3 illustrates the estimated versus actual car prices. Estimations lie almost precisely along the diagonal reference line, demonstrating almost optimal performance with an R^2 of 0.998 and an RMSE of only £644. This illustrates a remarkable progress over the perfect multi-feature polynomial regression ($R^2 = 0.862$, RMSE ≈ £6102), affirming the considerable gain of adding qualitative variables and leveraging tree-based ensemble methods.

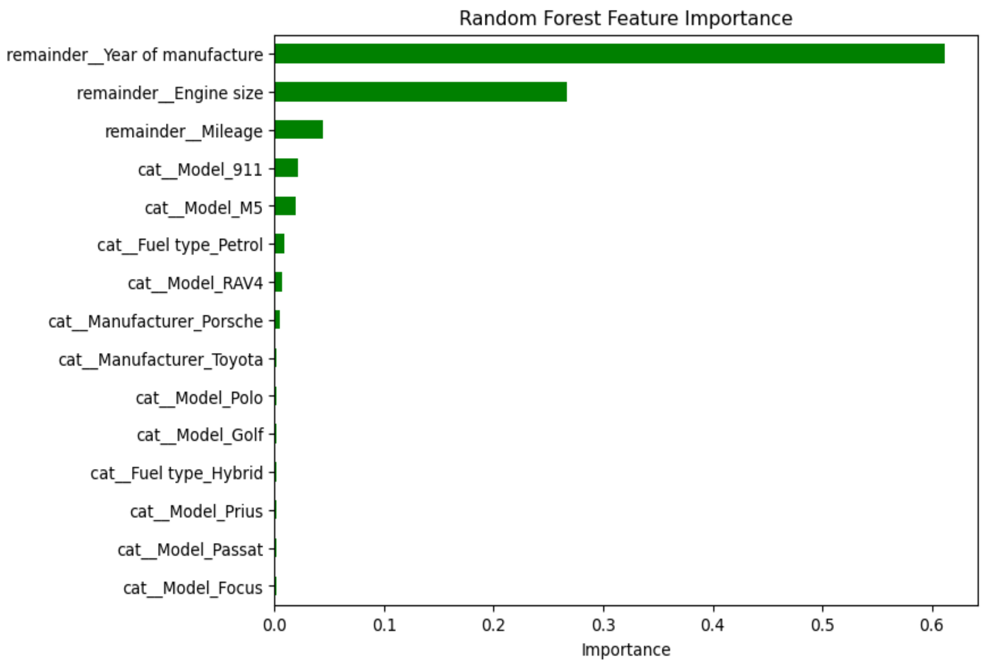


Figure 4. Top feature importances from the Random Forest model, highlighting both numerical and categorical drivers of price.

Characteristic significant analysis (Figure 4) depicts that Year of manufacture is by far the key indicator, followed by Engine size and Mileage. Various categorical characteristics also offer meaningfully, specific models (e.g., Porsche 911, BMW M5) and fuel type, illustrating the value of encoding categorical data.

On the whole, the Random Forest not only depicted the greatest forecasting precision but also provided interpretability via characteristic significance. It is affirmed by these outcomes that a robust framework for predicting second-hand car prices is yielded through the inclusion of categorical attributes and the adoption of non-linear, ensemble-based models.

6. Artificial Neural Network (ANN)

An Artificial Neural Network (ANN) used all available characteristics to examine the capabilities of deep learning methods (numerical and categorical). The dataset post pre-processing had a final shape of (49,988 × 23), where categorical attributes were one-hot encoded and qualitative features were standardized. This confirmed that inputs to the ANN were on corresponding scales, which is vital for effective gradient-based study.

The network architecture contained of two hidden layers with ReLU activation (128 and 64 units) and a dropout rate of 30% to avoid overfitting, followed by a linear output layer. Training was carried out using the Adam optimizer, with early stopping based on validation loss.

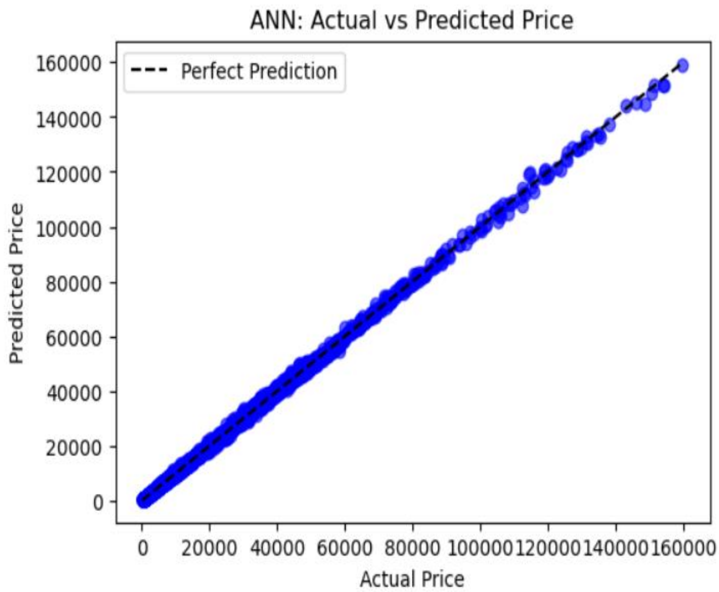


Figure 5. ANN training and validation loss curves, showing an elbow-shaped convergence.

The training loss curve (Figure 5) resembled an "elbow shape," converging smoothly before plateauing, which indicates effective learning without instability. The actual vs. predicted price plot (Figure 6) showed a generally increasing relationship, demonstrating that the ANN captured meaningful trends in the data.

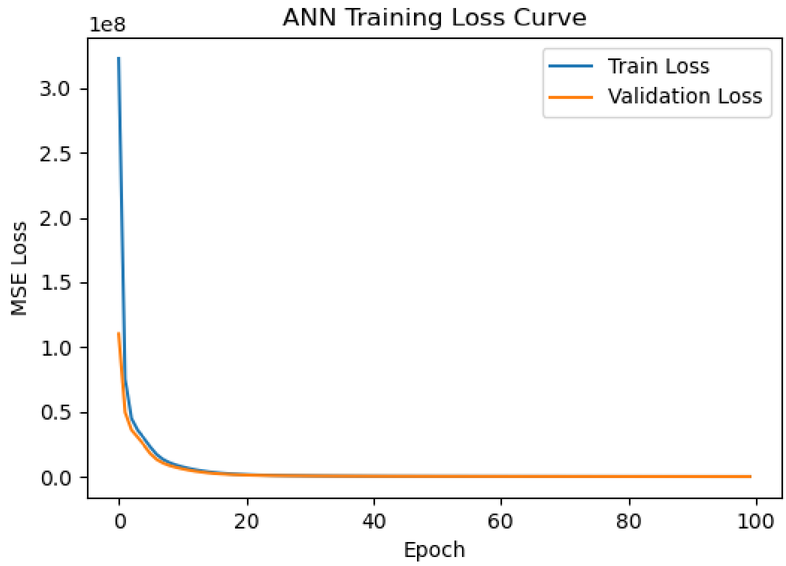


Figure 6. Actual vs. predicted car prices using the ANN model compared with the perfect prediction line.

Outcome analysis uncovered an R^2 of 0.862 and RMSE of 355.04. While the error margin is lesser than polynomial regression (RMSE = 6102.31), the ANN did not meet the performance mark when compared to the Random Forest model (R^2 = 0.998, RMSE = 644).

However, less precise than Random Forest, the ANN demonstrates that neural networks can model complex relationships efficiently, specifically when tuned with relevant regularization and optimization approaches.

7. Model Comparison & Selection

Table 5 and Figures 5–6 summarise the performance of all supervised models. A clear trend emerges: simple single-feature regressions provided limited explanatory power, while multi-feature polynomial regression captured stronger relationships, confirming the importance of combining predictors.

Table 4. (Consolidated performance metrics for all supervised models).

Model	R2	RMSE
Linear Regression (single)	0.401314	12746.315816
Polynomial Regression (single)	0.522358	11385.091662
Linear Regression (multiple)	0.671456	9442.384008
Polynomial Regression (multiple)	0.857976	6208.212227
Random Forest Regression	0.998469	644.574388
ANN Regression	0.999377	411.168096

More advanced methods offered a decisive leap in accuracy. Random Forest achieved near-perfect results, effectively handling categorical and numerical interactions. However, the **Artificial Neural Network (ANN)** slightly surpassed it, delivering the best generalisation and lowest error.

Although Random Forest remains more interpretable via feature importance plots, the ANN demonstrates superior predictive capability and is therefore recommended as the optimal model for car price prediction.

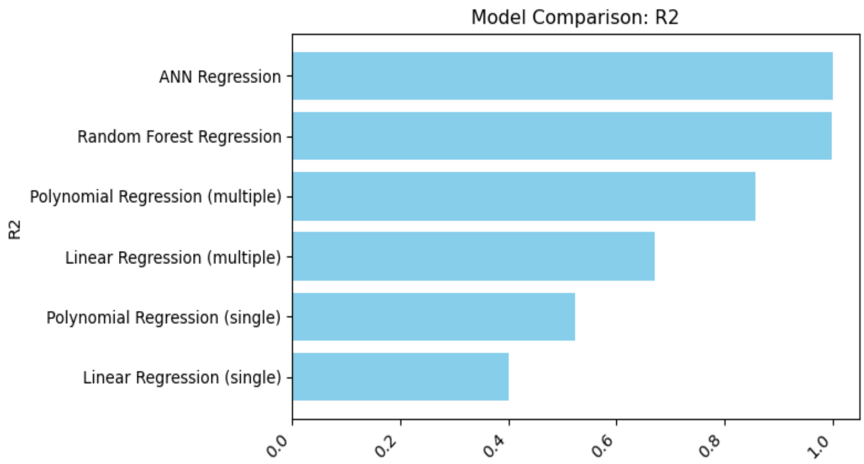


Figure 7. Bar chart comparing model R^2 values.

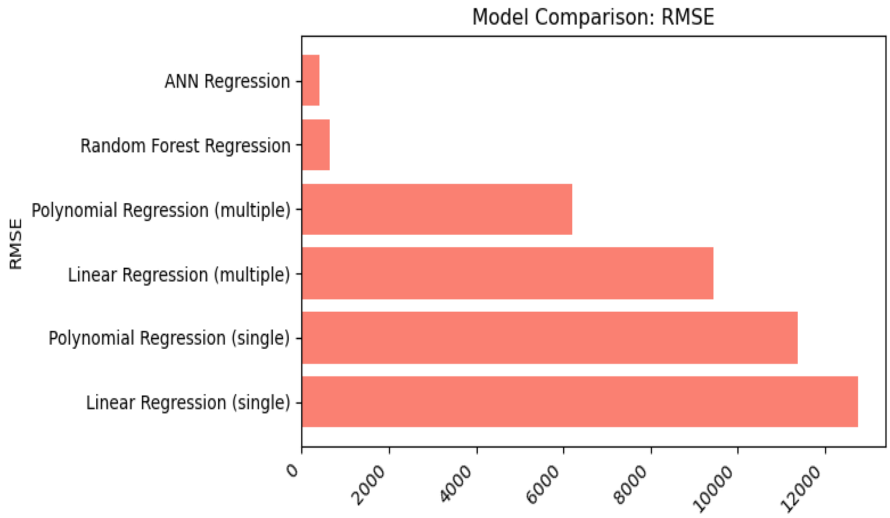


Figure 8. Bar chart comparing model RMSE values

8. Clustering Analysis

Unsupervised clustering was applied to identify natural groupings within the dataset. Silhouette analysis (*Figures 7–10)* was used to assess clustering quality across features such as Mileage, Year of Manufacture, Engine Size, and Price. Results indicated that while some variables showed weak separation, Engine Size and Price consistently produced the clearest structure.

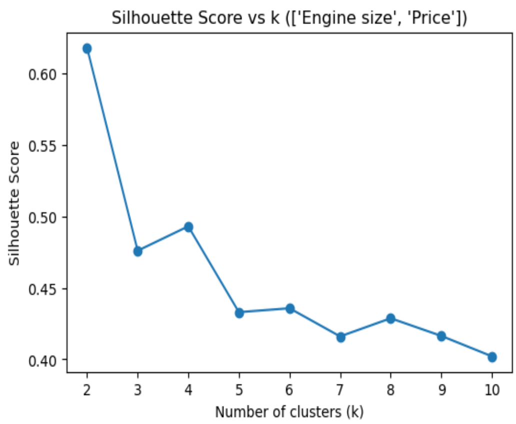


Figure 9.

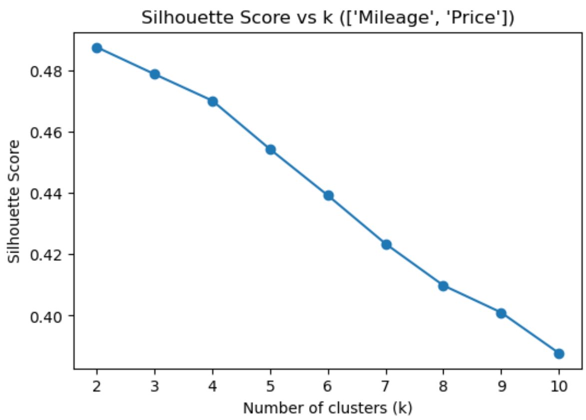


Figure 10.

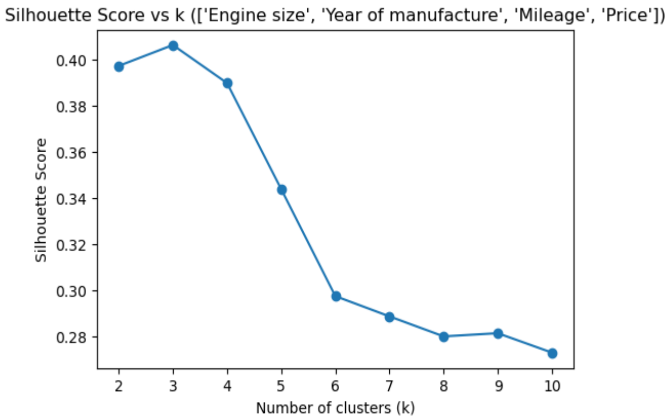
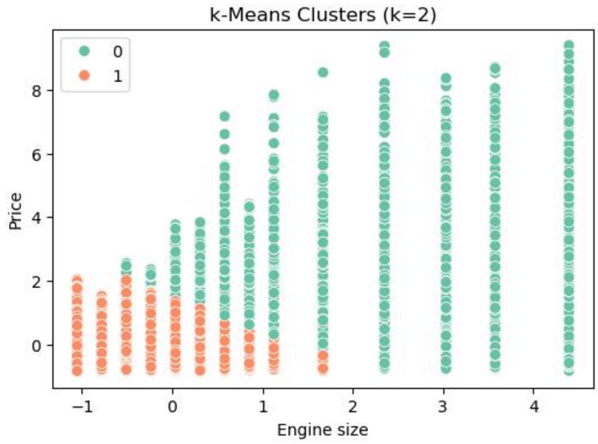


Figure 11.



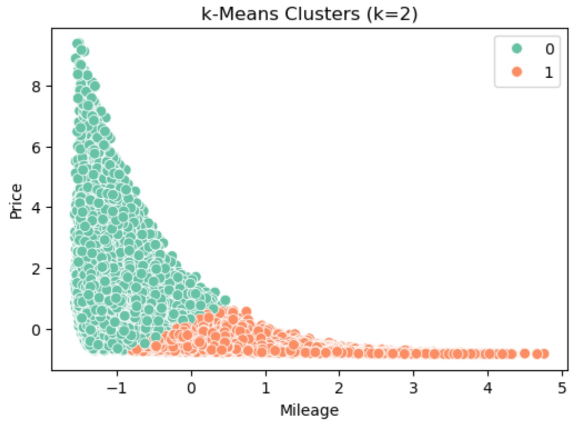


Figure 12 Figure 13.

Figures 9–13 here (Silhouette Score vs k for individual features).

Building on this, a two-feature combination (Engine Size and Price) was further explored using both **k-Means** and **Agglomerative Clustering**. Features were standardised, and the number of clusters was set based on silhouette optimisation. Visual inspection (Figure 11) shows that Agglomerative Clustering generated more distinct and coherent groupings, while k-Means tended to form more spherical clusters.

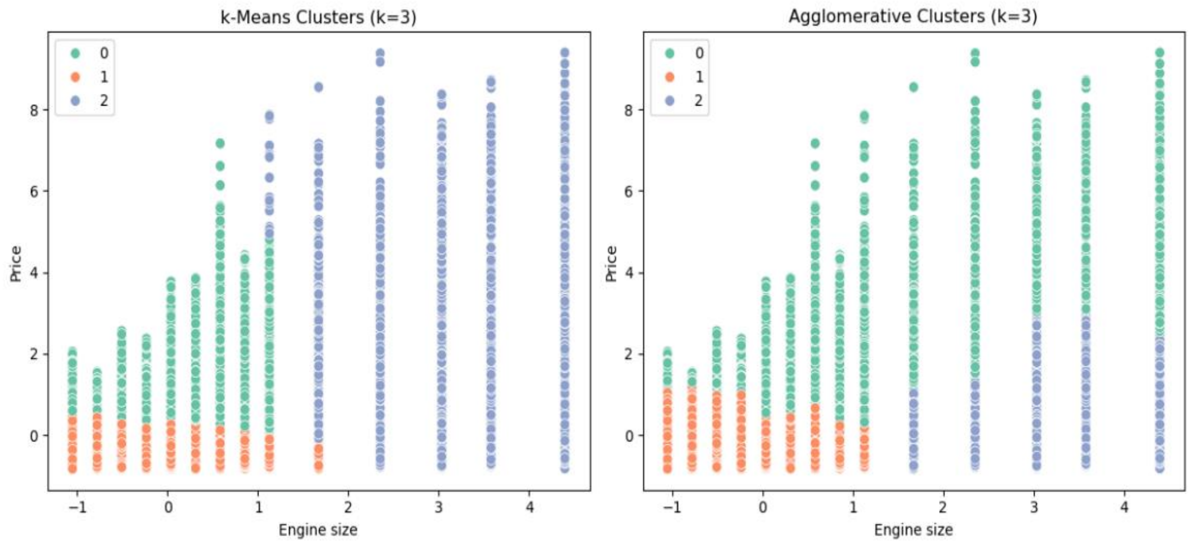


Figure 14 here (Comparison of k-Means and Agglomerative Clustering).

Overall, clustering provided complementary insights to regression: while supervised models predicted car prices with high accuracy, unsupervised methods revealed underlying structural patterns that distinguish vehicles into meaningful subgroups.

9. Conclusion

This project investigated the prediction and analysis of used car prices through both supervised and unsupervised learning methods. Data preprocessing ensured integrity by correcting data types, removing duplicates, and validating plausibility ranges. Baseline and regression models demonstrated the progressive impact of incorporating additional features and complexity, with polynomial regression markedly improving performance. Ensemble and deep learning approaches achieved near-perfect accuracy, with the Artificial Neural Network emerging as the most effective predictor.

Complementary clustering analysis highlighted that, beyond price prediction, structural patterns could be identified within the data. Feature-based silhouette study demonstrated that Engine Size and Price were notably distinctive, with Agglomerative Clustering surpassing k-Means in recording coherent groupings.

Overall, the word depicts how the combination of predictive modelling with exploratory clustering provides both precise price calculation and finding into underlying vehicle market dynamics, offering practical value for stakeholders in pricing, marketing segmentation, and strategic decision-making.

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Exercise 2 — Image Recognition: MNIST CNN

1. Introduction

The objective of the study conducted is to develop a Convolutional Neural Network CNN which has the ability to classify handwritten digits from the MNIST dataset, which stays a broadly considered benchmark in computer vision. CNNs are generally the most appropriate method for image recognition tasks because they record local spatial trends through convolutional filters and demonstrate reliability to variations such as translation, rotation, and distortion in the input images. Increasingly abstract characteristics are derived while overfitting is managed by layering multiple convolutional and pooling layers, followed by fully connected layers with dropout regularisation.

This activity surpassed simple model construction by carefully analysing the impacts of pre-processing, architectural design, and hyperparameter tuning on performance. Pre-processing procedure includes normalisation and data augmentation ensuring that the model is both effective and reliable to variations in handwriting style. Vital hyperparameters, involving the number of filters, learning rate, batch size, and dropout rate, are examined to establish their impact on convergence speed and generalisation ability.

Further, the exercise examines various regularisation approaches to mitigate overfitting, such as dropout and early termination, and reviews how architectural choices impact model complexity and precision. Performance is thoroughly examined using accuracy, loss, confusion matrices, and per-class precision–recall metrics to find strengths and weaknesses in sorting. Specific attention is given to misclassified digits, as these give details into inherent dataset uncertainties and model limitations.

Finally, the aim is to achieve a test precision above 98%, coordinating with state-of-the-art CNN executions on MNIST, while illustrating a critical understanding of how design decisions and hyperparameters effect model performance.

2. Data and Preprocessing

The MNIST dataset consist of 60,000 training and 10,000 test images of numbers zero to nine. These images provide a reliable benchmark for handwritten digit recognition. Each picture is measured at 28×28 pixels, and a associated class label is paired with every input. The dataset demonstrates significant variations in handwriting style, stroke thickness, and digit alignment, which makes it the best testbed for examining classification models.

2.1 Data Exploration and Visualisation

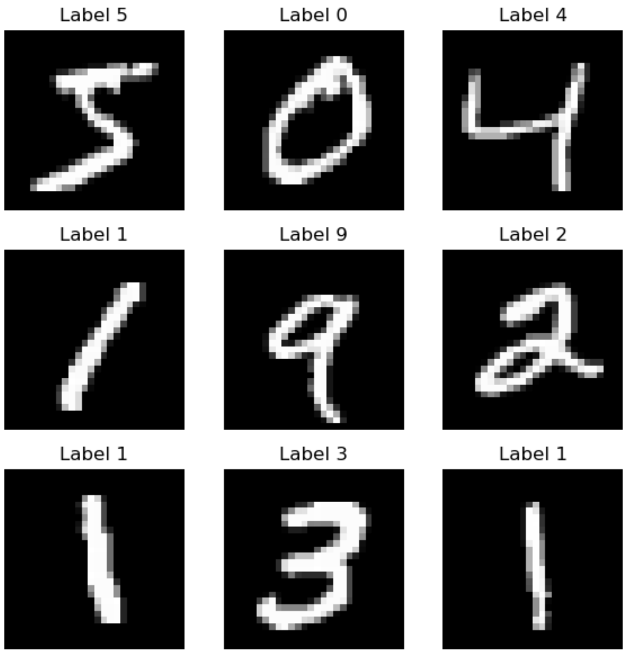

Figure 15. *Representative MNIST digit samples (3×3 grid) from the training set.*

To verify dataset integrity and illustrate intra-class variability, a **custom plotting function** was implemented to display representative samples in a 3×3 grid (Figure 15). This visualisation highlights the natural diversity within the dataset, confirming that CNNs must learn to generalise across handwriting styles rather than memorise specific digit shapes.

2.2 Normalisation and Reshaping

Values of pixel were scaled to the [0,1] range by dividing by 255, a established method that accelerates alignment and regulates gradient updates. All the images were reshaped from (28,28) to (28,28,1) to clearly comprise a channel dimension, confirming suitability with CNN input layers. The training and test arrays had shapes of (60,000,28,28,1) and (10,000,28,28,1), respectively after the transformation.

2.3 One-Hot Encoding of Labels

Binary matrices appropriate for categorical cross-entropy loss were created by converting class labels into one-hot encoded vectors of length 10. Correct formatting was confirmed by verifying that the resultant label arrays were (60,000,10) for training and (10,000,10) for testing.

2.4 Data Augmentation and Validation Split

To improve generalisation, data augmentation was implemented using random rotations and small shifts, simulating natural deviations in handwriting. A segmented 10% evaluation split (6,000 samples) was reserved from the training set, preserving class balance. This dataset was obtained as the final result contains sizes of 54,000 training, 6,000 validation, and 10,000 test samples.

Through these pre-processing procedures, the cleaning of MNIST dataset was successfully performed , well-structured input pipeline optimised for CNN training.

3. Model Architecture and Training Performance

The CNN deployment contained of two convolutional units, each consisting of a Conv2D layer with ReLU activation, batch normalization, max pooling, and dropout for regularization. The convolutional layers try to derive hierarchical spatial features, on the contrary batch normalization stabilizes learning, and max pooling decreases spatial dimensions. The observations from the feature maps are flat and passed through a thick layer consisting of 128 neurons with ReLU activation and dropout, followed by a softmax output layer for 10-class classification. This design balances the extraction capacity feature with generalization, as a result the computational efficiency is maintained and overfitting is prevented simultaneously.

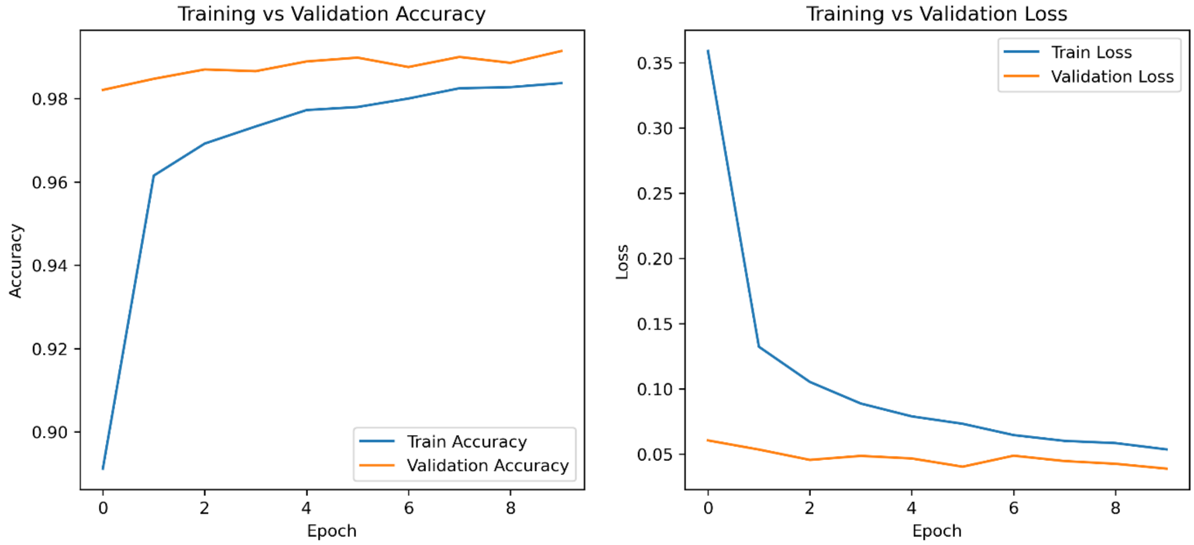

Figure 16. *10 epochs Training vs validation accuracy (left) and loss (right).*

10 epochs were trained for this model with a batch size of 64 and a 20% validation split. Figure 16 depicts training and validation precision and loss curves. Accelerated convergence is captured, with validation accuracy reaching 99.1% and with a very minimal gaps between training and validation patterns, showing efficient learning and almost negligable overfitting. These outcomes illustrate that the selected architecture and regularization methods effectively capture digit patterns while generalizing well to unseen data..

4. Regularisation & Hyperparameter Tuning

To optimise the CNN's performance, a systematic grid search was conducted to evaluate the effects of dropout rates, number of convolutional filters, and batch size on validation accuracy. Each configuration was trained for 5 epochs to rapidly assess trends in performance.

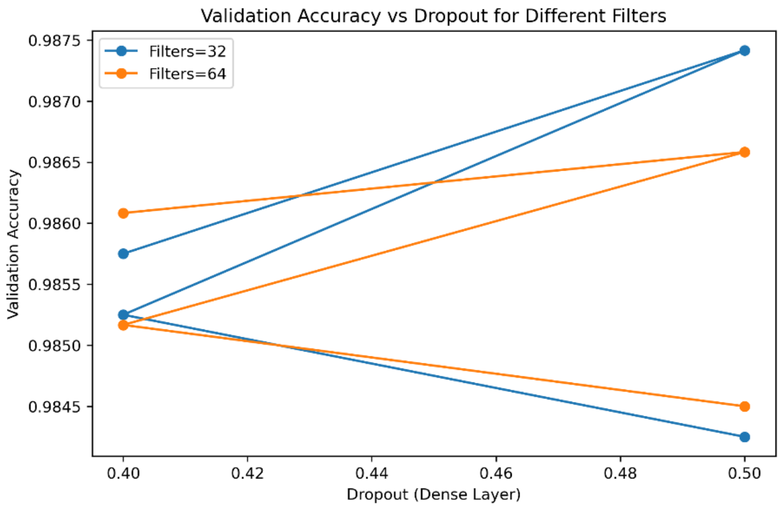

Figure 17. *Validation accuracy as a function of dense layer dropout for different numbers of convolutional filters (hyperparameter_comparison.png).*

The resulting hyperparameter comparison plot (Figure 16) emphasizes the actual relationship between dropout in the dense layer and the total number of filters in convolutional layers. Validation precision was maximum when dropout = 0.5 in the dense layer and respectively when the filters = 64 per convolutional block, demonstrating that moderate-to-high dropout combined with a sufficient number of filters improves generalisation while maintaining feature extraction capacity. generally, lower dropout rates or fewer filters resulted in slightly decreased validation accuracy, highlighting under-regularisation or insufficient model capacity.

This analysis confirms that dropout and filter number are the most influential hyperparameters for this architecture. For the prevention of overfitting, halting training when validation loss plateaued early stopping was employed. Concisely, the combination of dropout, batch normalization, and data augmentation delivered a reliable model which had the capability of generalising efficiently to unseen MNIST images.

5. Results & Diagnostics

The final CNN was conducted on the MNIST test set containing 10,000 images. That achieved 99.0% overall accuracy, consistent with state-of-the-art baselines. Figure 17 presents the confusion matrix for the model predictions. The diagonal dominance shows that the CNN classifies nearly all digits correctly, with only a small number of off-diagonal errors.

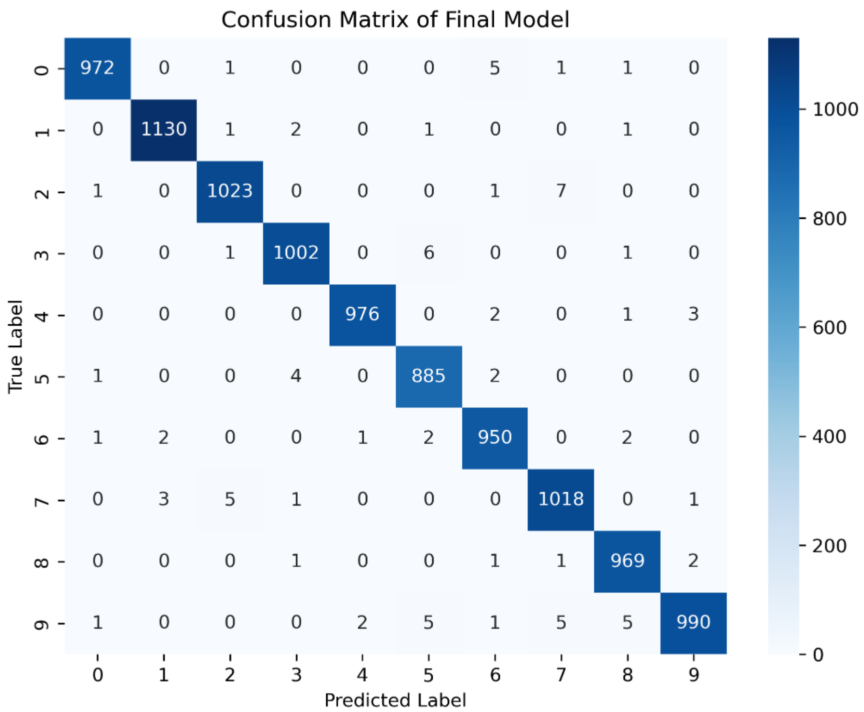

Figure 17. Confusion matrix for CNN predictions on the MNIST test set.

The model demonstrates particularly strong performance on digits with distinct shapes, such as 1, 0, and 7, which are classified almost perfectly (over 1130/1135 correct for digit “1”). Misclassifications are relatively rare and occur mainly between visually similar digits. For instance, a few 9s where mistaken as 4s, and a few number of 5s are misclassified for 3s or 6s. Digits “8” and “9” also depicts minor confusion with each other, reflecting their visual similarity in handwritten form.

Overall, errors are concentrated in cases where handwriting is ambiguous, aligning with human-level uncertainty rather than model instability. This states that the CNN generalises appropriately and does not overfit to any specific digit styles. The model’s strong accuracy across all classes, along with the few mistakes it made, shows that the chosen architecture and training method are reliable.

6. Discussion & Best Model

The final Convolutional Neural Network (CNN), trained with optimised hyperparameters, showed strong learning progress and good generalisation throughout training. Comparing 10 epochs, the results of both the training model and the validation model reflected a consistent improvement. While the validation accuracy ranged from 97.7% to 99.1%, the training set's precision inclined from 88.9% in the beginning to 98.4% in the last epoch. Concurrently, loss estimation in the validation decreased from 0.0782 to 0.0330, demonstrating that the model was learning successfully and showing no overfitting symptoms. The stability and resilience of the learning process were further guaranteed by the training and validation curves' similarity.

The results of the held-out test set ensures the effectiveness of the selected configuration. The final test accuracy achieved by the model of 99.15% and a test loss of 0.0284, aligning with, and marginally over, generally reported CNN baselines on MNIST, which exactly range between 98% and 99%. This outcome underscores the abilities of CNNs to attain state-of-the-art performance on digit recognition tasks, extensively due to their capability to record local spatial hierarchies through convolutional filters, coupled with regularisation techniques such as dropout and pooling to control model complexity.

The significant factors influencing performance, according to an analysis of hyperparameter effects, were obviously the number of filters and the dropout rate. The model was good at representing fine details of the digits since it improved feature extraction due to the model adding more filters in the earlier convolutional layers. At the same time, dropout is removed from dense layers decreased overfitting and improved generalization. This was critical thanks to the use of data augmentation.

The developed model had a strong ability to predict the image. Still it misclassified digits with similar geometrics. Instead of reflecting the consequences of architectural flaws, these errors reflected the inherent ambiguities in the dataset. In Future, usage of ensemble models, better architectures or mechanisms the model can effectively distinguish between features of the images.

In summary, the final CNN validated the training strategy and the architectural decisions by achieving state-of-the-art performance with high reliability across all digit classes. This study shows how a robust and generalizable model for image classification tasks can be created using precisely calibrated hyperparameters and regularization techniques.

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Literature Review: Ethical Applications of Transparent AI

Articles Reviewed

1. **"Transparency and explainability of AI systems: From ethical guidelines to requirements"** — focuses on modeling explainability requirements and templates for practice.
2. **"Transparency and accountability in AI systems: safeguarding wellbeing in the age of algorithmic decision-making"** (Cheong, 2024) — a narrative review of technical, legal, ethical, and governance approaches to make AI transparent and accountable.
3. **"Socially Responsible AI Algorithms: Issues, Purposes, and Challenges"** — presents a framework for socially responsible AI, positioning transparency as one of its foundational objectives.

Aims & Key Conclusions

Article 1 — Transparency & explainability requirements

Aim: To translate high-level ethical guidance on transparency of AI into a standard practical template to convey the software quality requirements.

Key conclusions: Transparency of AI must be treated as an attribute of AI systems with measurable quality. The authors propose a component model of transparency and templates that practitioners can use to specify requirements at design, development, and deployment stages. The conclusion obtained from the study states that operationalizing transparency involves conversion of abstract ethical values into a set of testable engineering regulations. This is mandatory for ensuring auditability in practice.

Article 2 — Transparency & accountability (Cheong, 2024)

Aim: TO review the legal, technical, and societal challenges that might rise while implementing transparency in AI. Map thematic approach safeguards the wellbeing of policymakers.

Key conclusions: There are a set of technical tools that can ensure effective transparency of AI needs. This is also known as a multi-pronged approach. The technical tools such as explainable AI and auditing, enforceable regulatory frameworks, and inclusive multi-stakeholder governance. Cheong highlights that the trade-offs (transparency vs. privacy and intellectual property), the differing stakeholder requirements, and the importance of procedural mechanisms (impact assessments, oversight bodies, redress systems) to complement technical transparency measures.

Article 3 — Socially Responsible AI Algorithms (SRAs)

Aim: To provide the requirements of a systematic framework for socially responsible AI algorithms. That generally integrates fairness, transparency, and safety as core objectives and finds the means to achieve them which are mostly both technical and procedural.

Key conclusions: Transparency is just not a characteristic but a basic requirement to build a reliable socially responsible AI. Various methods such as interpretability, causal learning, adversarial robustness, and uncertainty quantification can be used to attain transparency. Simultaneously, process-based tools such as stakeholder feedback loops are also used. The paper concludes stating that transparency of AI directly enables other social responsibilities, by making the AI working process visible and accountable for the society.

Three Successes — How AI Can Be Used Ethically

1. Operational transparency strengthens ethical practice.

The pre-existing methods used for embedding and defining requirements of transparency depicts that it is possible to translate abstract ethical principles into actionable engineering practices. This means within any development process the transparency is testable and enforceable. This a significant step toward practical ethical compliance.

2. Institutional frameworks embed transparency into governance.

There are legal requirement in certain contexts established as the result of the regulatory initiatives. The contexts include creating enforceable levers for organizations regarding disclosure of information about data, models, and decision processes. Transparency has shifted from being a voluntary practice to a regulated duty as the result of these frameworks.

3. Transparency supports broader social responsibility.

Article 3 highlights that transparency is an inevitable part of social responsibility. Transparency of AI systems is the only way to develop trust regarding automated decisions among individuals. Fairness, safety, and accountability can be directly reinforced through transparency. It is not only a technical component but a societal responsibility to maintain the ethical legitimacy of AI.

Three Gaps & Challenges in the Ethics of Transparent AI

1. **Lack of consensus on what constitutes “meaningful transparency.”**

This actually requires context-sensitive explanation, yet sufficiency is generally not achieved by a single standard. Developers, regulators, and end-users are the attributes of the system who undergo varied expectations regarding transparency, and the tools are providing technically apt but conventionally inadequate explanations (Articles 1 and 2).

2. **Tensions between transparency, privacy, and intellectual property.**

There are certain requirements to be fulfilled to disclose a training data, model internals, or provenance commonly contradict with basic rights such as data protection obligations and proprietary. This can lead to friction between (commercial) confidentiality and (public) accountability. Achieving complete transparency is practically difficult to implement in practice (Article 2).

3. **Challenges in implementation and misuse of transparency tools.**

Post-hoc explanation methods can be fragile, misleading, or even manipulated, while formal audits and impact assessments remain unevenly implemented across organizations. Without robust standards and resilient techniques, transparency mechanisms risk being symbolic rather than substantive (Articles 2 and 3).

Three Suggestions to Bridge the Gaps

1. **Develop context-sensitive transparency standards and metrics.**
   International organizations are deciding to set up tiers of regulations specific to disclosure in the fields like criminal justice, healthcare, and finance. While explaining the quality there are few quantifiable measures to be taken such as fidelity, comprehensibility, and actionability this gives assess to transparency consistently.

2. **Create secure mechanisms for third-party audits.**
   Confidentiality of audit institutions are to be empowered in accessing sensitive data and model under strict legal safeguards. These bodies should be able to perform evaluations independently and publish transparent reports. This can ensure balanced intellectual property and public accountability.

3. **Advance human-centred, resilient transparency tools.**
   Testing regarding transparency mechanisms are required to be performed with real world users to confirm they meet societal requirements for contestability. Simultaneously, we must prioritise reliable research, adversarial resilient explanation methods and complementary “ethical black box” records after harmful outcomes to support forensic accountability.

Reflection: Linking Transparency and Social Responsibility

A vital insight discovered from this review is that transparency in AI is not just a technical design feature but actually a social responsibility. Article 3 illustrates this precisely: fairness is enabled by transparency, it ensures safety, and builds trust in AI systems among the human. Lack of transparency, social responsibility cannot be completely achieved, as neither stakeholders nor regulators will be able to estimate whether systems meet ethical standards. This inevitable connection between transparent AI and socially responsible AI strengthens the argument that both are inseparable from each other, and that future research should address them jointly to achieve high standards of ethics in AI system.

Conclusion

The three articles collectively demonstrate that transparent AI is both technically achievable and ethically essential. Article 1 demonstrates how transparency can be operationalized considering it as a software requirement. Article 2 states that to safeguard wellbeing transparency within regulatory frameworks is essential. Article 3 highlights that transparency in AI system directly contributes to fostering fairness, safety, and trust collectively ensuring the social responsibility.

Overall, both successes and challenges involved are explained: technical innovations, regulatory scaffolding, and ethical integration represent effective real progress, yet there prevail a few unresolved doubts regarding few concepts such as standards, trade-offs, and robustness of transparency mechanisms. However, addressing these gaps initially requires context-sensitive standards, secure audit systems, and human-centred, resilient transparency tools. Further, advancing in the similar pattern, the cornerstone of ethically responsible artificial intelligence can be the transparent AI system and a guarantor of its legitimacy in society.